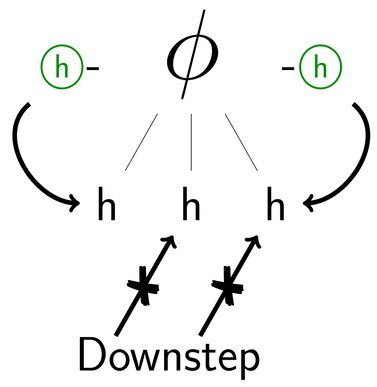
Downstep Blocking by Register Circumfixes

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1. Main Claim

- Downstep blocking in interrogative utterances can be modelled as circumfixation of h-register features.
- Problem: Local left-to-right downstep.
  - Global blocking in a large domain.
  - No simple concatenative solution.
- Solution: h-register circumfix overwrites intervening high registers to fulfill CONTIGUITY constraint.
- Downstep would create new violations of this constraint.

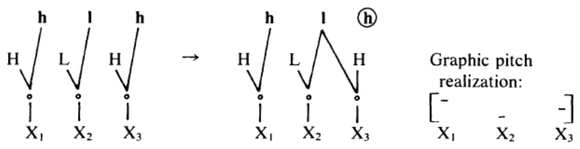
(1) Register overwriting blocks downstep



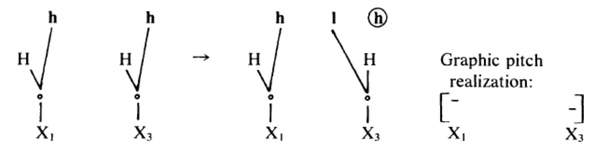
2. Data: Downstep Blocking

- Automatic downstep: global pitch lowering triggered by low tone on following high tone.
- Non-automatic downstep: global pitch lowering on high tone following high tone.

(2) Automatic downstep (Snider, 1990)



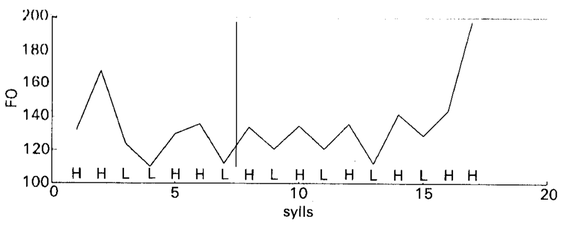
(3) Non-automatic downstep (cf. Snider 1990)



Hausa

(4) Hausa polar question pitch track

(Inkelas & Leben, 1990, 22-23)

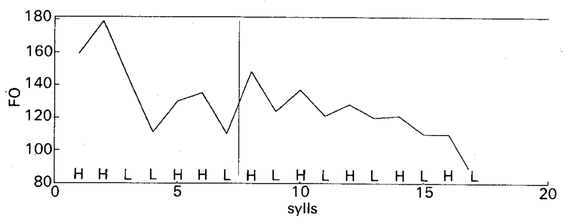


Question: ... ϕ ϕfinal — DS no DS

Declarative: ... ϕ ϕfinal — DS DS

(5) Hausa declarative question pitch track

(Inkelas & Leben, 1990, 22-23)

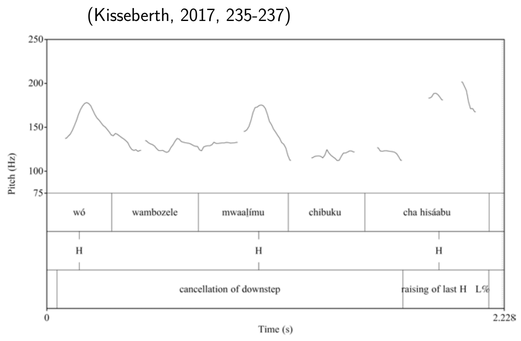


- In Hausa, low tones induce downstep on following high tone in declarative sentences.
- Automatic downstep blocked in last phrase of a questions. (cf. Inkelas & Leben 1990).

Chiminii

(6) Chimiini polar questions pitch track

(Kisseberth, 2017, 235-237)

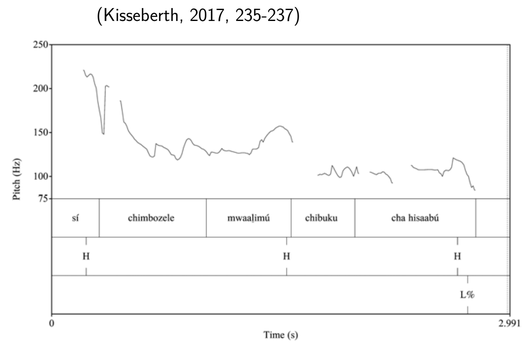


Question: ... ϕ ϕfinal — no DS no DS

Declarative: ... ϕ ϕfinal — DS DS

(7) Chimiini declarative pitch track

(Kisseberth, 2017, 235-237)



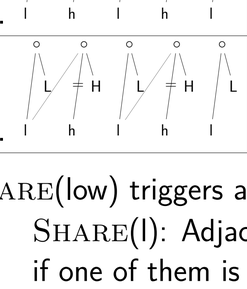
- In Chimiini, privative high tones induce a downstep if in sequence.
- This automatic downstep is blocked in the whole utterance in questions. (cf. Kisseberth 2017).

Selected references: •Inkelas, Sharon & William Leben. 1990. Where phonology and phonetics intersect: the case of Hausa intonation. In *Papers in laboratory phonology i: Between the grammar and physics of speech*, 17–34. Cambridge, UK: Cambridge University Press. •Kisseberth, Charles. 2017. Chimiini intonation. In Laura Downing & Annie Rialland (eds.), *Intonation in African tone languages*, 225–284. Berlin: De Gruyter Mouton. •Revithiadou, Anthi. 2007. Colored turbid accents and containment: A case study from lexical stress. In Silvya Blaho, Patrik Bye & Martin Krämer (eds.), *Freedom of analysis?*, 149–174. Berlin: Mouton de Gruyter. •Snider, Keith L. 1990. Tonal upstep in Krachi: Evidence for a register tier. *Language* 66(3). 453–474. •Trommer, Jochen. 2011. *Phonological aspects of Western Nilotic mutation morphology*: Leipzig: University of Leipzig dissertation. •Yu, Alan. 2007. *A natural history of infixation*. Oxford: Oxford University Press.

3. Analysis: Register Circumfixes

- High register circumfix (ⓗ- -ⓗ) attaches to domain of downstep supression.

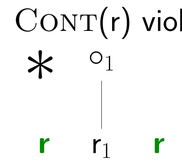
(8) Question Syntax



- Pivot Theory of Affixation (Yu 2007): Affixes specified for prosodic constituent and position they attach to (cf. boundary tones).
- Register Tier Theory ((Snider, 1990)): Autosegmental division between tones and registers. Both linked to tonal root nodes (○).
  - Register features h and l lower/raise pitch globally.
  - Tone features H and L lower/raise pitch locally.
- Contiguity constraint forces register circumfix to overwrite intervening h-registers.
- Association of low register (=downstep) is blocked, since tonal root nodes are occupied.

(9) CONTIGUITY(r): Count one violation for every tonal root node $\circ_1$, such that $\circ_1$ is visibly associated to a register feature r_1 of morphological affiliation c_1 that intervenes between two register features of the same morphological affiliation c_2. (cf. Trommer 2011)

(10) CONT(r) violation



- Colored Containment (Revithiadou 2007): Constraints refer:
  - to surface (visible) structure and/or complete structure.
  - to identity or non-identity of morphological affiliation.

Hausa

(11) Automatic downstep in Hausa
Evaluation of [...ròn ↓álà↓rámmà]

Input:	CONT(r)	SH(l)
a.		∗∗∗!∗
☞ b.		∗∗

(13) Downstep blocking
Derivation of [...ròn álàrámmà?]

Input:	CONT(r)	SH(l)
a.	∗∗∗∗!∗	∗∗∗∗
☞b.	∗∗∗	∗∗∗∗
c.	∗∗∗∗!∗	∗∗

- SHARE(low) triggers automatic downstep.

(12) SHARE(l): Adjacent ○ must share l if one of them is associated to l

- In questions, high register cirumfix attaches to last prosodic phrase of the utterance.
- CONTIGUITY forces circumfix to overwrite intervening high registers.
- Automatic downstep is blocked, since low register spreading would keep the high register cirumfix from docking – a violation of CONTIGUITY.

Chimiini

(14) Non-Automatic downstep in Chimiini
Evaluation of a declarative sentence

Input:	CONT(r)	OCP(h)
a.		∗!∗
☞ b.		

(15) Downstep blocking in Chimiini
Evaluation of a question

Input:	CONT(r)	OCP(h)
a.	∗!∗∗	∗∗∗∗
☞b.		∗∗∗∗
c.	∗!∗∗	∗∗

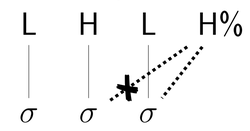
- Chimiini downstep consists of epenthetic low registers.
- Inserted due to OCP(high) constraint.
- In questions, downstep deletion – again – due to ⓗ- -ⓗ attached to whole utterance.
- These are forced to overwrite intervening registers due to CONTIGUITY(register).
- Downstep is blocked again, since inserting and associating low registers would intervene between the ⓗ- -ⓗ.
- This would fatally violate CONTIGUITY(r).

4. Discussion

Boundary tones

- Simple single boundary tone not sufficient.
- Locality: can either link only to target (stressed, leftmost, rightmost) and thus have a local effect or to all underspecified tone bearing units (i.a. Hyman 1990).
- In Hausa, underspecification does not work, because of falling tones.

(16) Boundary tone locality



Alternative downstep representations

- Phonetic implementation of HH (Odden, 1982)
  - Modularity: Phonetics would refer to morphosyntactic features.
  - Non-local application of blocking.
  - Interaction with other phonetic processes (cf. Inkelas & Leben 1990)
- Floating low tone (Clements & Ford, 1979)
  - Declarative floating low tone insertion.
  - Question floating low tone deletion.
  - Abstract processes not affecting any associated tone.

Cophonologies/Indexed Constraints

- Requires a rule/constraint that applies only in question and blocks downstep only in the last phonological phrase (cf. (Inkelas & Leben, 1990))
- Implies several phonological grammars.
- Constraints make reference to final phrase.
- Prediction: regular phonological downstep blocked only in last phonological phrase in some language.

Rule-based alternative

- With the same information present, a rule could achieve a similar result by deleting downstep in the presence of the floating register circumfix.

(17) Downstep Deletion Rule

$[-h] \rightarrow [+h] / \mathbf{[+h]} \ldots _, +H \ldots \mathbf{[+h]}$

- Duke-of-York Derivation: earlier downstep is reversed later in derivation.
- Diacritic features: high register features stay floating and do not actually dock.
- Locality: Non-local context specification.